QoS Questions

7 questions | Original English questions and source answers

Question 1

Which two actions are performed by the Weighted Random Early Detection mechanism? (Choose two)

Options

A. It supports protocol discovery

B. It guarantees the delivery of high-priority packets

C. It drops lower-priority packets before it drops higher-priority packets

D. It can identify different flows with a high level of granularity

E. It can mitigate congestion by preventing the queue from filling up

Answer

C. It drops lower-priority packets before it drops higher-priority packets

E. It can mitigate congestion by preventing the queue from filling up

Source explanation

Weighted Random Early Detection (WRED) is just a congestion avoidance mechanism. WRED drops packets selectively based on IP precedence. Edge routers assign IP precedences to packets as they enter the network. When a packet arrives, the following events occur:

1. The average queue size is calculated.
2. If the average is less than the minimum queue threshold, the arriving packet is queued.
3. If the average is between the minimum queue threshold for that type of traffic and the maximum threshold for the interface, the packet is either dropped or queued, depending on the packet drop probability for that type of traffic.
4. If the average queue size is greater than the maximum threshold, the packet is dropped.

WRED reduces the chances of tail drop (when the queue is full, the packet is dropped) by selectively dropping packets when the output interface begins to show signs of congestion (thus it can mitigate congestion by preventing the queue from filling up). By dropping some packets early rather than waiting until the queue is full, WRED avoids dropping large numbers of packets at once and minimizes the chances of global synchronization. Thus, WRED allows the transmission line to be used fully at all times.

WRED generally drops packets selectively based on IP precedence. Packets with a higher IP precedence are less likely to be dropped than packets with a lower precedence. Thus, the higher the priority of a packet, the higher the probability that the packet will be delivered (-> answer 'It drops lower-priority packets before it drops higher-priority packets' is correct).

Reference: https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/qos_conavd/configuration/15-mt/qos-conavd-15-mt-book/qos-conavd-cfg-wred.html

Source: CCNA 200-301 Questions.html | Question 1

Question 2

Which two QoS tools can provide congestion management? (Choose two)

Options

A. PQ

B. CBWFQ

C. PBR

D. CAR

E. FRTS

Answer

A. PQ

B. CBWFQ

Source explanation

This module discusses the types of queueing and queueing-related features (such as bandwidth management) which constitute the congestion management QoS features:

Class-based WFQ (CBWFQ): extends the standard WFQ functionality to provide support for user-defined traffic classes. For CBWFQ, you define traffic classes based on match criteria including protocols, access control lists (ACLs), and input interfaces. Packets satisfying the match criteria for a class constitute the traffic for that class.

Priority queueing (PQ): With PQ, packets belonging to one priority class of traffic are sent before all lower priority traffic to ensure timely delivery of those packets.

Reference: https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/qos_conmgt/configuration/xe-3s/qos-conmgt-xe-3s-book/qos-conmgt-oview.html

Note: Committed Access Rate (CAR) is only used for bandwidth limitation by dropping excessive traffic.

Source: CCNA 200-301 Questions.html | Question 2

Question 3

Which feature can you implement to reserve bandwidth for VoIP calls across the call path?

Options

A. PQ

B. CBWFQ

C. RSVP

D. round robin

Answer

C. RSVP

Source explanation

The Resource Reservation Protocol (RSVP) protocol allows applications to reserve bandwidth for their data flows. It is used by a host, on the behalf of an application data flow, to request a specific amount of bandwidth from the network. RSVP is also used by the routers to forward bandwidth reservation requests.

Source: CCNA 200-301 Questions.html | Question 3

Question 4

What does traffic shaping do to reduce congestion in a network?

Options

A. drops packets

B. queues without buffering packets

C. buffers without queuing packets

D. buffers and queues packets

Answer

D. buffers and queues packets

Source explanation

The following diagram illustrates the key difference between traffic policing and traffic shaping. Traffic policing propagates bursts. When the traffic rate reaches the configured maximum rate (or committed information rate), excess traffic is dropped (or remarked). The result is an output rate that appears as a saw-tooth with crests and troughs. In contrast to policing, traffic shaping retains excess packets in a queue and then schedules the excess for later transmission over increments of time. The result of traffic shaping is a smoothed packet output rate.
Note: Committed information rate (CIR): The minimum guaranteed data transfer rate agreed to by the routing device.

Source: CCNA 200-301 Questions.html | Question 4

Question 5

Which option is the main function of congestion management?

Options

A. providing long-term storage of buffered data

B. discarding excess traffic

C. classifying traffic

D. queuing traffic based on priority

Answer

D. queuing traffic based on priority

Source: CCNA 200-301 Questions.html | Question 5

Question 6

Which feature or protocol determines whether the QoS on the network is sufficient to support IP services?

Options

A. LLDP

B. EEM

C. IPSLA

D. CDP

Answer

C. IPSLA

Source explanation

IP SLA allows an IT professional to collect information about network performance in real time. Therefore it helps determine whether the QoS on the network is sufficient for IP services or not.

Cisco IOS Embedded Event Manager (EEM) is a powerful and flexible subsystem that provides real-time network event detection and onboard automation. It gives you the ability to adapt the behavior of your network devices to align with your business needs.

Source: CCNA 200-301 Questions.html | Question 6

Question 7

What is the purpose of traffic shaping?

Options

A. to provide fair queuing for buffered flows

B. be a marking mechanism that identifies different flows

C. to limit the bandwidth that a flow can use

D. to mitigate delays over slow links

Answer

C. to limit the bandwidth that a flow can use

Source explanation

The primary reasons you would use traffic shaping are to control access to available bandwidth, to ensure that traffic conforms to the policies established for it, and to regulate the flow of traffic in order to avoid congestion that can occur when the sent traffic exceeds the access speed of its remote, target interface.

Source: CCNA 200-301 Questions.html | Question 7